<!-- Sync Impact Report:

- Version change: 0.0.0 → 1.0.0
- Modified principles: N/A (initial concrete version)
- Added sections:
  - I. Conventional Commits (NON-NEGOTIABLE)
  - II. Commit Scope and Trailers (NON-NEGOTIABLE)
  - III. Project Metrics and COCOMO (NON-NEGOTIABLE)
  - Additional Constraints: Branching/Review
  - Development Workflow: Commit/Review Gates
- Removed sections: None
- Templates requiring updates: None (project-specific)
- Follow-up TODOs: None -->

Proyecto Constitution

Core Principles

I. Conventional Commits (NON-NEGOTIABLE)

Every commit MUST follow the Conventional Commits structure:

```
<tipo>(<alcance opcional>): <descripción corta en imperativo>

[cuerpo opcional explicativo]

[pie de página opcional]
```

Allowed types: `feat`, `fix`, `docs`, `style`, `refactor`, `test`, `chore`. Descriptions MUST use imperative present tense in English or Spanish as appropriate, start with lowercase (unless proper nouns/siglas), have no trailing period, and be under 50 characters. If a body is needed, separate with a blank line and wrap lines to a maximum of 72 characters.

II. Commit Scope and Trailers (NON-NEGOTIABLE)

- **Scope is mandatory for feature-related work:** Commits MUST include the functionality number as scope (e.g., `feat(001): ...`, `fix(001): ...`).
- **Deletion/Rewrite trailer required:** Commits that delete or rewrite code MUST include the trailer `Causa:` with one of `requisito|defecto|refactor` (e.g., `Causa: refactor`).
- **Issue references:** When applicable, use footers such as `Closes #<id>`, `Fixes #<id>`, or `Refs #<id>` to link commits to tracked work.

III. Project Metrics and COCOMO (NON-NEGOTIABLE)

- **Scale factors:** The project scale factors MUST be qualified in the row `000-proyecto` of `metricas/factores-cocomo.csv`.
- **Effort multipliers:** Effort multipliers MUST be qualified at the end of each `/speckit-plan`.

System Context

TraductorLSEC Mobile App

- **Purpose:** Enable a person who does not know Ecuadorian Sign Language (LSEC) to understand a deaf person who does, by sending sign videos captured by the mobile device.
- **Video input:** Videos of signs with a maximum duration of 10 minutes, streamed in constant packets to avoid unnecessary device storage consumption.
- **Camera:** The app requires full camera access for capture.
- **Processing:** Videos are processed by an external deep learning model deployed in the cloud. Each request MUST include the clip duration and a unique ID. The model returns real-time sign translations as text.
- **Chat/history:** Translations are shown in a WhatsApp-style chat with history of up to the last 30 translations.
- **TTS:** A text-to-speech component on the mobile client generates audio from the translations, with the required audio permissions.
- **UX:** The app must be highly intuitive and easy to use; error handling is key to avoid user confusion or misunderstandings.
- **Connectivity:** The app must work under low connectivity conditions.
- **Authentication:** No login is required; no data is stored in the cloud, only the remote model is used for processing.

Additional Constraints

Branching and Review

- **Atomic commits:** Each commit MUST represent a single logical change.
- **Staged review:** Before committing, changes MUST be reviewed in the staging area (e.g., via Fork's Commit panel) to ensure only intended files are included.
- **No secrets:** Secrets, keys, or credentials MUST NOT be committed.

Development Workflow

Commit and Review Gates

- **Message validation:** Commit messages MUST conform to Principles I and II.

- **Plan completion gate:** Before finalizing a `/speckit-plan`, COCOMO effort multipliers MUST be recorded as required by Principle III.
- **Metrics maintenance:** `metricas/factores-cocomo.csv` MUST remain up to date with the current project state.

Architecture

Server and Client Integration

- **Server deployment:** The server is deployed to connect with the mobile client, sending JSON packets to it in order to obtain the translation.
- **MVC structure:** The server code MUST follow an MVC folder structure.
- **TDD:** Development MUST follow Test Driven Development across all phases before any commit.
- **Connection config:** Connection is configured through a `.env` file shipped alongside the compiled build; the endpoint MUST NOT be exposed in source code.
- **Language:** All code MUST be written in Spanish.

Amendment Procedure

Amendments to this Constitution require:

1. A documented proposal describing the change and rationale.
2. Review and approval by the project team.
3. Update of the Constitution version according to the versioning policy below.

Versioning Policy

- **MAJOR:** Backward incompatible governance/principle removals or redefinitions.
- **MINOR:** New principle/section added or materially expanded guidance.
- **PATCH:** Clarifications, wording, typo fixes, non-semantic refinements.

Current Version: 1.2.0
Ratified: 2026-10-20
Last Amended: 2026-10-21

Compliance Review

All commits, plans, and process changes MUST comply with this Constitution. Deviations require explicit justification and team approval.